9/2/26.

LISTS

```
l=["Charvi", "Ara", "Siddhi", "Khushi", "Vasu"]
print(l)
l.append("boo")
print(l)
l.extend(["hoo", "loo"])
print(l)
l.pop(2)
print(l)

for i in l:
    print(i)

for i in range(1,6):
    print(l[i])
```

Output

```
['Charvi', 'Ara', 'Siddhi', 'Khushi', 'Vasu']
['Charvi', 'Ara', 'Siddhi', 'Khushi', 'Vasu', 'boo']
['Charvi', 'Ara', 'Siddhi', 'Khushi', 'Vasu', 'boo', 'hoo', 'loo']
['Charvi', 'Ara', 'Khushi', 'Vasu', 'boo', 'hoo', 'loo']
Charvi
Ara
Khushi
Vasu
boo
hoo
loo
```

```
Ara
Khushi
Vasu
boo
hoo
```

NEOPIXEL using lists

```
from machine import Pin
import neopixel
import time
l=[(255,0,0), (255,0,0) ..... (255,0,0)]   x16
n= neopixel.NeoPixel(Pin(12),16)
for i in range (0,16):
    n[i]= l[x]
    time.sleep(0.1)
    n.write()
```

NEOPIXEL using Lists and Random

```
from machine import Pin
import neopixel
import random
import time
l = [(255,0,0), ..... (255,0,0)]   x16
n= neopixel.NeoPixel(Pin(12),16)
while True:
    for i in range(0,16):
        x= random.randint(0,15)
        n[i] = l[x]
        time.sleep(0.1)
        n.write()
```

NEOPIXEL quadrant random colour.

```
from machine import Pin
import neopixel
import random
import time
n= neopixel.NeoPixel(Pin(12),16)
while True:
    r= random.randint(0,255)
    g= random.randint(0,255)
    b= random.randint(0,255)
    r1=          "
    g1=          "
    b1=          "
    r2=          "
    g2=          "
    b2=          "
    r3=          "
    g3=          "
    b3= random.randint(0,255)
    for i in range(0,4):
        n[i] = (r,g,b)
        print(r,g,b)
        time.sleep(0.1)
        n.write()
    for j in range(4,8):
        n[j] = (r1,g1,b1)
```

```
    print(r1, g1, b1)
    time.sleep(0.1)
    n.write()
for k in range(8,12):
    n[k] = (r2, g2, b2)
    print(r2, g2, b2)
    time.sleep(0.1)
    n.write()
for l in range(12,16):
    n[l] = (r3, g3, b3)
    print(r3, g3, b3)
    time.sleep(0.1)
    n.write()
```

TO CREATE A RANDO COLOUR (in general)

```
from machine import Pin
import neopixel
import time
import random
r = random.randint(0,255)
g = random.randint(0,255)
b = random.randint(0,255)
print(r, g, b)
```

only one colour will be generated for a run time.

```
n = neopixel.NeoPixel(Pin(12), 16))
while True:
    for i in range(0,16):
        n[i] = (r, g, b)
        time.sleep(0.2)
        n.write()
```

if,

Each loop will get new colour

```
while True:
    r = random.randint(0,255)
    g = random.randint(0,255)
    b = random.randint(0,255)
```

DIFFERENCE B/W "If" & "Elif"

```
x = 5
if x>0:
    print('ok')
elif x==5:
    print('okk')
elif x<10:
    print('okkk')
```

output

ok

'If' has higher priority over 'elif'. Even if 'If' statement is true, elif statements will not be executed even if they're true.

```
if x>0:
    print('ok')
if x==5:
    print('okk')
if x<0:
    print('okkk')
else:
    print('no')
```

OUTPUT

ok
okk
no.

else statement considers only the latest 'If' statement

All 'If' statements have equal priority ∴ even if first 'If' is true, the rest will still be executed.

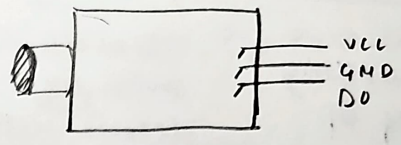
```
if x<0:
    print('ok')
elif x==5:
    print('okk')
elif x>0:
    print('okkk')
else:
    print('no')
```

OUTPUT

okk

∵ 'If' cond^n is wrong, 'Elif' is executed, but, only the first elif is executed despite the other being true.

LDR (LIGHT DEPENDENT RESISTOR)



VCC → 3v3
GND (sgm501) → GND
DO → GPIO

] ESP32

```
from machine import Pin
import time
ldr = Pin(14, Pin.IN, Pin.PULL_UP)
while True:
    ldr_v = ldr.value()
    print(ldr_v)
    time.sleep(0.2)
```

OUTPUT is 0 in general, 1 when you cover with hand (no light)

Difference b/w LDR & IR sensors

LDR: function according to intensity of light.

(↓) resistance as light intensity (↑)

IR: uses IR LED to emit light and detects reflected IR light thru obstacle.

If you keep a torch infront of IR sensor, and switch it off

↳ IR: value remains 0 as torch as an object acts as an obstacle.

↳ LDR: displays 0, no light source.

WHEN u PUSH BUTTON, COUNTER gets INCREMENTED

```
from machine import Pin
import time
c = 0
s = Pin(4, Pin.IN, Pin.PULL_UP)
while True:
    s_val = s.value()
    if s_val == 0:
        c += 1
        print(c)
        time.sleep(0.5)
```

OUTPUT

1
2
3
.
.
.

WHEN counter reaches '5', it resets to 0

```
from machine import Pin
import time
c = 0
s = Pin(4, Pin.IN, Pin.PULL_UP)
while True:
    s_val = s.value()
    if s_val == 0:
        c += 1
        print(c)
        time.sleep(0.5)
        if c == 5:
            c = 0
            print(c)
```

OUTPUT

1
2
3
4
5
0
1
.
.
.

WHEN YOU push button, each LED of NEOPIXEL lights up.

(X)

```
from machine import Pin
import time
import neopixel
n = neopixel.NeoPixel(Pin(12), 16)
s = Pin(4, Pin.IN, Pin.PULL_UP)
i = 0
while True:
    s_val = s.value()
    if s_val == 0:
        i += 1
        time.sleep(0.5)
        print(i)
        n[i] = (255,0,0)
        time.sleep(0.1)
        n.write()
```

Here, we observed that led no. '0' did not light up coz of increment.

∴ CORRECT METHODS (✓)

(1)

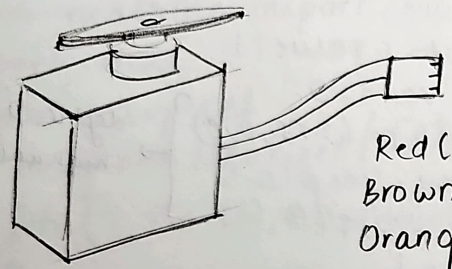
```
i = 0
while True:
    s_val = s.value()
    if s_val == 0:
        n[0] = (255,0,0)
        time.sleep(0.1)
        n.write()
        i += 1
        time.sleep(0.5)
        print(i)
        n[i] = (255,0,0)
        time.sleep(0.1)
        n.write()
```

define '0' LED manually.

OR

(2) put i = -1 ☺

MICRO SERVO MOTOR



Red (+5V)
Brown (GND)
Orange (PWM) → GPIO

Servo	ESP32	Power Module
+5v (red)		+5v
GND (brown)	GND	GND (-)
PWM (orange)	GPIO	

PWM for servo motor

Pulse width to be given to servo	Duty Cycle	Value (*) for micropy code	position (*) (degree) A°
0.5ms	2.5%	35 to 45	0
1.5ms	7.5%	77	90
2.5 ms	12.5%	110 to 125	180

(*) Keep constant

$\boxed{\text{Frequency} = 50\ Hz}$

Time period = 20ms

Code (Basic)

```
from machine import Pin, PWM
import time
p = PWM(Pin(4))          # Pin.OUT optional ∵ default.
p.freq(50)
while True:
    p.duty(40)           # 0°
    time.sleep(0.1)
    p.duty(77)           # 90°
    time.sleep(0.1)
```

] motor moves back & forth.

SMOOTH movement of motor

```
from machine import Pin, PWM
import time
p = PWM(Pin(4))
p.freq(50)
while True:
```

```
for i in range(0,120,1):
    p.duty(i)
    time.sleep(0.01)
for j in range(120,-1,-1):
    p.duty(j)
    time.sleep(0.01)
```

WEEK 6.

16/2/26